# Bacteria–Phage–Antibiotic Coevolution

v0.4 | Theoretical matrix model | Research summary | 7 November 2026


## Abstract

This report summarizes the v0.4 status of a researcher-configurable theoretical matrix model for studying antibiotic resistance, bacteriophage effects and changes in bacterial populations. The model is not biologically calibrated and must not be used for clinical decisions, laboratory procedures or treatment-efficacy prediction.


## Research question and approach

Antibiotic susceptibility and susceptibility to productive infection in a specified phage-host pairing are treated as separate dimensions. Pre-existing resistance, new mutations, selection and possible gene transfer are conceptually distinguished. Combined antibiotic-phage effects are not presumed beneficial or harmful.

## Six bacterial states

S: susceptible to the specified antibiotic; I: intermediate/transitional phenotype; R: resistant. F+: susceptible to productive infection in the defined phage-host pairing; F−: not productively infected under the same conditions. State order: S/F+, S/F−, I/F+, I/F−, R/F+, R/F−. Categories are disjoint under fixed measurement conditions; the intermediate category does not double-count a cell.

## Mathematical core

$n(t+1) = G(t)\,T(t)\,n(t)$. n is a six-element absolute-population vector; T is a nonnegative 6×6 transition matrix whose columns sum to one; G is a diagonal matrix of six independent nonnegative population multipliers. Transition precedes multiplication. Transition flow $Q(i,j,t) = T(i,j,t)\,n(j,t)$; with reference volume V, transition density $q = Q/V$. When the total is positive, relative frequency $x(i)=n(i)/\mathrm{sum}(n)$; frequencies are undefined at extinction.

## Transition hypotheses

Both S ↔ I ↔ R directions can be examined as researcher-defined hypotheses with or without antibiotics. Phage-associated phenotype changes can also be explored in both directions, but mutation, selection, phenotypic switching and horizontal gene transfer must not be conflated. Phages cannot transform into bacteria. Phage-mediated gene transfer is a separate optional mechanism.

## Uncertainty principle

An uncharacterized variant or transition is classified as 'uncertain', not 'neutral'. Missing coefficients are not silently converted to zero; documented alternative matrices may be evaluated with explicit assumptions, provenance, units, context and uncertainty ranges. Mechanism weights between 0 and 1 scale model contributions; they are not biological probabilities.

## Scenarios and independence

Researchers can independently vary initial populations, transition matrices and population multipliers. Distinct matrices may represent no antibiotic/no phage, antibiotic only, phage only, and combined conditions. Order of operations can affect outcomes. The current six-state implementation does not dynamically reproduce phage populations.

## Version history and differences

v0.1: abstract two-strategy frequency game. v0.2: three strategies and 20 abstract scenarios. v0.3: numeric population and molarity inputs recorded as metadata only; they do not drive dynamics. v0.4: six states, column-stochastic transition matrix, independent population multipliers, transition flows and optional density outputs. The v0.4 software is a theoretical computation engine, not a validated biological model.

## Python implementation

GitHub files: models/v04.py, tests/test_v04.py, docs/v04_python.md and docs/v04_design.md. The code rejects missing/NaN entries, negative entries and transition columns that do not sum to one. It computes flows, post-transition populations, post-multiplier populations and relative frequencies. Tests have been authored; this report does not claim execution in every environment.

## Limitations and next steps

Real infection cycles, dynamic phage replication, antibiotic pharmacodynamics, pathogen-specific parameters, environmental effects and empirical validation are absent. Matrix coefficients may not be identifiable from observations alone; identifiability, sensitivity, numerical convergence and independent data comparison are required. Added external mutagens and ionizing radiation are outside the baseline, but natural mutation is not assumed zero.

## Conclusion

The principal contribution is a researcher-configurable six-state matrix architecture that separates antibiotic phenotype from phage susceptibility. Outputs are mathematical scenario comparisons, not experimental evidence or treatment recommendations.

Repository: github.com/altugkuran/phage-antibiotic-coevolution